

CLEWs and OG-PHL: Simulating the Philippine Energy Plan 2023–2050

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Recap

CLEWs: What it is, what it has

- Partial equilibrium
- Inputs: Prices, demands, basic resource, precip., temp.
- Technology: Production, cost function, inputs and outputs
- Outputs: Allocations, final costs, resource use, emissions, particulates

OG-PHL: What it is, what it has

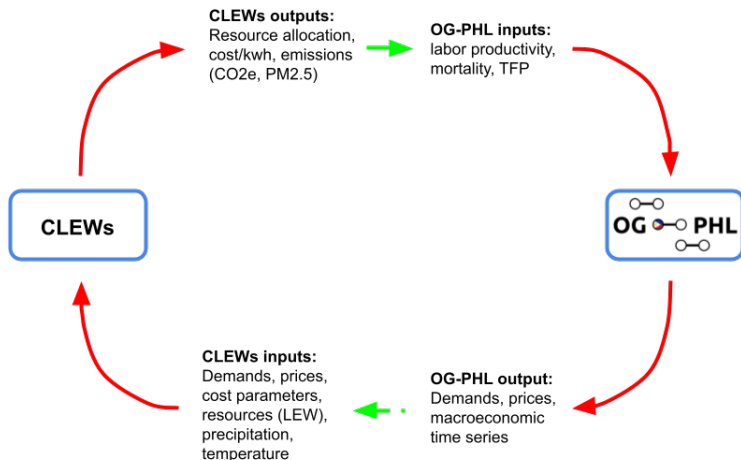
- General equilibrium
- Inputs: household, firm, government parameters
- Technology: utility functions, production functions
- Outputs: prices, allocations, microeconomic and macroeconomic time series

Philippine Energy Plan 2023–2050

<https://legacy.doe.gov.ph/pep>

- “...attain an inclusive and **equitable economic growth**... through the provision of secure, sustainable, and resilient energy strategies.”
- “...energy sector’s goal to ... attaining **clean energy** future.”
- 35% and 50% renewable energy share in power generation by 2030 and 2040
- 10% EV penetration for road transport (motorcycles, cars, jeepneys) by 2040
- Resumption of indigenous oil and gas exploration
- Introduction of liquefied natural gas (LNG) portfolio
- Establishment of strategic petroleum reserves
- Exploration of Hydrogen’s potential

CLEWS and OG-PHL Flow Diagram



A Comment on OG-PHL/CLEWs Validity

- Calibration is never done
- Calibration of OG-PHL and CLEWs is “pretty good” right now
- The delta's (changes) produced by the models are credible

This is an advanced proof of concept

To really make OG-CLEWs operational, more work needs to be done.